Steps toward reproducible research

Karl Broman

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Slides: kbroman.org/Talk_RepoRes

```
Karl -- this is very interesting,
however you used an old version of
the data (n=143 rather than n=226).

I'm really sorry you did all that
work on the incomplete dataset.

Bruce
```

The results in Table 1 don’t seem to correspond to those in Figure 2.

Where did we get this data file?

Why did I omit those samples?

Which image goes with which experiment?

How did I make that figure?

In what order do I run these scripts?

“Your script is now giving an error.”

“The attached is similar to the code we used.”

Reproducible research

organize the data and code in a way
that you can hand them to someone else
and they can re-run the code
and get the same results
(the same figures and tables)

Reproducible

vs.

Replicable

Reproducible

vs.

Correct

kbroman.org/steps2rr

A little bit reproducible
is better than not reproducible.

A little bit open
is better than not open.

Strive to make each project
a bit better organized than the last.

Organize your project

File organization and naming
are powerful weapons against chaos.

– Jenny Bryan

Organize your project

Your closest collaborator is you six months ago,
but you don't reply to emails.

(paraphrasing Mark Holder)

Organize your project

Have sympathy for your future self.

Organize your project

```
RawData/                Notes/
DerivedData/            Refs/

Python/                 ReadMe.txt
R/                      ToDo.txt
Ruby/                   Makefile

Analysis/
Figures/
```

Chaos

```
AimeeNullSims/     Deuterium/             Ping/
AimeeResults/      ExtractData4Gary/      Ping2/
AnnotationFiles/   FromAimee/             Ping3/
Brian/             GoldStandard/          Ping4/
Chr6_extrageno/    HumanGWAS/             Play/
Chr6_segdis/       Insulin/               Prdm9/
ChrisPlaisier/     Int2_for_Mark/         RBM_PlasmaUrine_2012-03-08/
Code4Aimee/        Islet_2011-05/         Slco1a6/
CompAnnot/         MappingProbes/         StudyLineupMethods/
CondScans/         MultiProbes/           kidney_chr6.R
D2O_2012-02-14/    NewMap/                pck2_sucla2.R
D2O_cellcycle/     Notes/                 penalties.txt
D2Ocorr/           NullSims/              transeQTL4Lude/
Data4Aimee/        NullSims_2009-09-10/
Data4Tram/         PepIns_2012-02-09/
```

Choose good names for things

```
betw_tissue_corr.R      expr_scatterplot_allprobes.R  gve_similarity_alltissues.R
coatcolor_lod.R         expr_scatterplots_dup.R       gve_similarity.R
colors.R                expr_scatterplots_mix.R       gve_supp.R
cover_fig.R             expr_scatterplots_swap.R      insulin_lod.R
eqtl_counts_10.R        expr_swaps.R                  local_eqtl_locations.R
eqtl_counts.R           func.R                        my_plot_map.R
eve_hist.R              genotype_plates.R             my_plot_scanone.R
eve_scheme.R            gve_hist.R                    sex_vs_X.R
eve_similarity.R        gve_new.R                     xchr_fig.R
eve_similarity_supp.R   gve.R                         xist_and_y.R
expr_corr_dup.R         gve_scheme.R
expr_corr_mix.R         gve_similarity_2ndbest.R
```

Choose good names for things

```
fig1.png          fig5.png
fig10.png         fig6.png
fig2.png          fig7.png
fig3.png          fig8.png
fig4.png          fig9.png
```

Choose good names for things

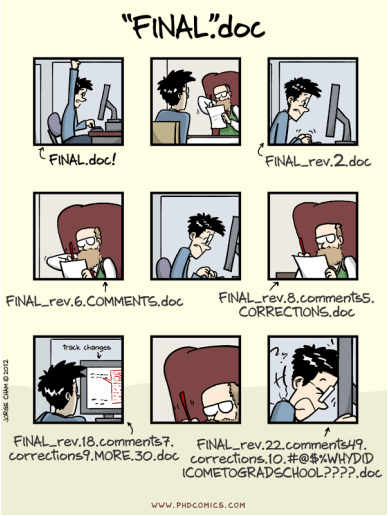
- Machine readable
  - No spaces
  - No special characters except _ and -
- Human readable
  - Explain the contents
- Consistent
  - Name similar files in a similar way
- Make use of computer's sorting
  - pad numbers with 0's (e.g., `01`, `02`, ...)
  - start with general grouping, then more specific
  - dates like `2019-05-14`

xkcd.com/1179

Choose good names for things

```
0_vcf2db.R
1_prep_geno.R
2_prep_pheno_clin.R
2_prep_pheno_otu.R
3_prep_covar.R
4_prep_analysis_pheno_clin.R
4_prep_analysis_pheno_otu.R
5_scans.R
6_grab_peaks.R
7_find_nearby_peaks.R
```

No “final” in file names



No “final” in file names

```
Deprecated/
ReadMe.txt
adipose_int1_final.RData
adipose_int2_final.RData
adipose_mlratio_final.RData
adipose_mlratio_nqrank_final.RData
adipose_prcomp.RData
aligned_geno_with_pmap.RData
batches_final.RData
batches_raw_final.RData
cpl_final.RData
d2o_final.RData
gastroc_int1_final.RData
gastroc_int2_final.RData
gastroc_mlratio_final.RData
gastroc_mlratio_nqrank_final.RData
gastroc_prcomp.RData
hypo_int1_final.RData
hypo_int2_final.RData
hypo_mlratio_final.RData
hypo_mlratio_final_old.RData
hypo_mlratio_nqrank_final.RData
hypo_mlratio_nqrank_final_old.RData
hypo_omit.RData
hypo_prcomp.RData
islet_int1_final.RData
islet_int2_final.RData
islet_mlratio_final.RData
islet_mlratio_nqrank_final.RData
islet_prcomp.RData
kidney_int1_final.RData
kidney_int2_final.RData
kidney_mlratio_final.RData
kidney_mlratio_nqrank_final.RData
kidney_prcomp.RData
lipomics_final_rev2.RData
liverTG_final.RData
liver_int1_final.RData
liver_int2_final.RData
liver_mlratio_final.RData
liver_mlratio_nqrank_final.RData
liver_prcomp.RData
mirna_final.RData
necropsy_final_rev2.RData
plasmaurine_final_rev.RData
pmark.RData
rbm_final.RData
```

No “final” in file names

```
Deprecated/
ReadMe.txt
adipose_int1_final.RData
adipose_int2_final.RData
adipose_mlratio_final.RData
adipose_mlratio_nqrank_final.RData
adipose_prcomp.RData
aligned_geno_with_pmap.RData
batches_final.RData
batches_raw_final.RData
cpl_final.RData
d2o_final.RData
gastroc_int1_final.RData
gastroc_int2_final.RData
gastroc_mlratio_final.RData
gastroc_mlratio_nqrank_final.RData
gastroc_prcomp.RData
hypo_int1_final.RData
hypo_int2_final.RData
hypo_mlratio_final.RData
hypo_mlratio_final_old.RData
hypo_mlratio_nqrank_final.RData
hypo_mlratio_nqrank_final_old.RData
hypo_omit.RData
hypo_prcomp.RData
islet_int1_final.RData
islet_int2_final.RData
islet_mlratio_final.RData
islet_mlratio_nqrank_final.RData
islet_prcomp.RData
kidney_int1_final.RData
kidney_int2_final.RData
kidney_mlratio_final.RData
kidney_mlratio_nqrank_final.RData
kidney_prcomp.RData
lipomics_final_rev2.RData
liverTG_final.RData
liver_int1_final.RData
liver_int2_final.RData
liver_mlratio_final.RData
liver_mlratio_nqrank_final.RData
liver_prcomp.RData
mirna_final.RData
necropsy_final_rev2.RData
plasmaurine_final_rev.RData
pmark.RData
rbm_final.RData
```

Choose good names for things

```
batches_raw_v1.rds
batches_v1.rds
clinical_cpl_v2.rds
clinical_d2o_v2.rds
clinical_lipomics_v4.rds
clinical_liverTG_v2.rds
clinical_mirna_v2.rds
clinical_necropsy_v4.rds
clinical_plasmaurine_v3.rds
clinical_rbm_v2.rds
Deprecated/
geneexpr_int1_adipose_v2.rds
geneexpr_int1_gastroc_v2.rds
geneexpr_int1_hypo_v2.rds
geneexpr_int1_islet_v2.rds
geneexpr_int1_kidney_v2.rds
geneexpr_int1_liver_v2.rds
geneexpr_int2_adipose_v2.rds
geneexpr_int2_gastroc_v2.rds
geneexpr_int2_hypo_v2.rds
geneexpr_int2_islet_v2.rds
geneexpr_int2_kidney_v2.rds
geneexpr_int2_liver_v2.rds
geneexpr_mlratio_adipose_v2.rds
geneexpr_mlratio_gastroc_v2.rds
geneexpr_mlratio_hypo_v1.rds
geneexpr_mlratio_hypo_v2.rds
geneexpr_mlratio_islet_v2.rds
geneexpr_mlratio_kidney_v2.rds
geneexpr_mlratio_liver_v2.rds
geneexpr_mlratio_nqrank_adipose_v2.rds
geneexpr_mlratio_nqrank_gastroc_v2.rds
geneexpr_mlratio_nqrank_hypo_v1.rds
geneexpr_mlratio_nqrank_hypo_v2.rds
geneexpr_mlratio_nqrank_islet_v2.rds
geneexpr_mlratio_nqrank_kidney_v2.rds
geneexpr_mlratio_nqrank_liver_v2.rds
geneexpr_omit_hypo.rds
geneexpr_prcomp_adipose_v2.rds
geneexpr_prcomp_gastroc_v2.rds
geneexpr_prcomp_hypo_v2.rds
geneexpr_prcomp_islet_v2.rds
geneexpr_prcomp_kidney_v2.rds
geneexpr_prcomp_liver_v2.rds
geno_aligned_w_pmap.rds
geno_pmark.rds
ReadMe.txt
```

Document your work

- What is all of this stuff?
- What was your analysis process?

→ `ReadMe` files

Organizing data in spreadsheets

	A	B	C	D	E	F	G
1	1MIN						
2			Normal			Mutant	
3	B6	146.6	138.6	155.6	166	179.3	186.9
4	BTBR	245.7	240	243.1	177.8	171.6	188.1
5							
6	5MIN						
7			Normal			Mutant	
8	B6	333.6	353.6	408.8	450.6	474.4	423.8
9	BTBR	514.4	610.6	597.9	412.1	447.4	446.5

Organizing data in spreadsheets

	A	B	C	D
1	ttt_min	strain	mutation	response
2	1	B6	normal	146.6
3	1	B6	normal	138.6
4	1	B6	normal	155.6
5	1	B6	mutant	166
6	1	B6	mutant	179.3
7	1	B6	mutant	186.9
8	1	BTBR	normal	245.7
9	1	BTBR	normal	240
10	1	BTBR	normal	243.1
11	1	BTBR	mutant	177.8
12	1	BTBR	mutant	171.6
13	1	BTBR	mutant	188.1
14	5	B6	normal	333.6
15	5	B6	normal	353.6

Organizing data in spreadsheets

- Make it a rectangle
- Individual measurements as rows; variables as columns
- Single header row
- One item per cell
- No empty cells
- No calculations in the raw data
- No highlighting or coloring as data

Broman and Woo (2018) Am Stat 72:2-10

doi.org/gdz6cm

“What the heck is ‘`FAD_NAD SI 8.3_3.3G`’?”

Metadata

- Create a data dictionary
  - Explain each column
  - Include different versions of the variable names (compact vs descriptive)
  - Units
  - Allowable values
- The metadata are data
  - Make it a rectangle

Data dictionary

	A	B	C	D
1	name	plot_name	group	description
2	mouse	Mouse	demographic	Animal identifier
3	sex	Sex	demographic	Male (M) or Female (F)
4	sac_date	Date of sac	demographic	Date mouse was sacrificed
5	partial_inflation	Partial inflation	clinical	Indicates if mouse showed partial pancreatic inflation
6	coat_color	Coat color	demographic	Coat color, by visual inspection
7	crumblers	Crumblers	clinical	Indicates if mouse stored food in their bedding
8	diet_days	Days on diet	clinical	Number of days on high−fat diet

Everything with a script

If you do something once,
you'll do it 1000 times.

Small corrections

	A	B	C	D
1	id	Rt Kidney wt	Rt Adipose wt	Liver wt
2	DO−121	294	757	930
3	DO−122	296	583	439
4	DO−123	NA	834	527
5	DO−124	513	808	600
6	DO−125	381	780	493
7	DO−126	225	1.066	355
8	DO−127	262	1.03	512
9	DO−128	231	0.687	497
10	DO−129	263	0.932	580
11	DO−130	266	985	906

Small corrections

	A	B	C	D
1	id	Rt Kidney wt	Rt Adipose wt	Liver wt
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8	DO−127	262	1.03	512
9	DO−128	231	0.687	497
10	DO−129	263	0.932	580
11	DO−130	266	985	906

Differing column names

	A	B	C	D	E
1	id	glucose.mg.dl.0	glucose.mg.dl.5	glucose.mg.dl.15	glucose.mg.dl.30
2	DO−121	99.165552	349.303552	286.092208	312.047704
3					
4					
5					
6					
7					
8					
9					
10					
11					

	A	B	C	D	E
1	id	glucose.0	glucose.5	glucose.15	glucose.30
2	DO−221	145.742786	206.452638	216.640608	299.55501
3	DO−222	138.010378	342.866944	339.836676	276.148802
4	DO−223	138.219362	407.443	336.858654	235.501414
5	DO−224	100.445504	310.944638	384.97722	308.907044
6	DO−225	121.030428	290.41196	345.740474	313.818168
7	DO−226	118.418128	189.524934	159.692468	144.488882
8	DO−227	117.4777	395.321928	448.612848	310.369932
9	DO−228	98.773632	149.452252	245.637138	317.423142
10	DO−229	122.44107	260.63174	231.008258	202.272958

Differing column order

	A	B	C	D	E
1	id	glucose.mg.dl.0	glucose.mg.dl.5	glucose.mg.dl.15	glucose.mg.dl.30
2	DO−121	99.165552	349.303552	286.092208	312.047704

	A	B	C	D	E
1	id	glucose.0	glucose.5	glucose.15	glucose.30
2	DO−221	145.742786	206.452638	216.640608	299.55501

	A	B	C	D	E
1	id	glucose.0	insulin.0	glucose.5	insulin.5
2	DO−321	66.839405	0.04	246.685995	0.04
3	DO−322	98.12509	0.51185	246.25574	1.4062
4	DO−323	94.68305	1.7812	448.1068	1.0248
5	DO−324	121.051535	0.0882	407.355505	0.63475
6	DO−325	122.95695	0.19155	298.193665	0.6467
7	DO−326	201.447755	0.7454	386.51887	0.6081

Metadata solution

	A	B	C	D	E
1	short_name	file	from_column	id_column	column_offset
2	mouse	wave2_sheet1.csv	mouse #	mouse	0
3	sex	wave2_sheet1.csv	sex	mouse	0
4	sac_date	wave2_sheet1.csv	sac date	mouse	0
5	num_islets	ex_vivo_waves1−3.csv	# islets	mouse	0
6	Ins_per_islet	ex_vivo_waves1−3.csv	IC	mouse	0
7	Glu_0min	gtt2.csv	glucose.mg.dl.0	id	0
8	Ins_0min	gtt2.csv	insulin.ng.ml.0	id	0
9	Glu_tAUC	gtt2.csv	glucose.mg.dl.tAUC	id	0
10	Glu_iAUC	gtt2.csv	glucose.mg.dl.iAUC	id	0
11	Ins_tAUC	gtt2.csv	insulin.ng.ml.tAUC	id	0

“In what form would you like the data?”

“In what form would you like the data?”

The answer should always be

“In its present form.”

Reproducible reports

Gough project diagnostics

Karl Broman, 3 March 2014

Combine genotypes and phenotypes

I've combined the initial genotypes (using the re-clustered genotypes for plates 14-16) with the well-behaved portion of the re-run genotypes. I'm focusing on 36813 markers that are informative (though, as we'll see, there are still a lot of badly behaved and basically non-informative markers that need to be removed). I've combined data on replicate samples, to give one set of genotype calls for each sample.

There are 1497 genotyped mice and 1464 phenotyped mice. All of the mice in the phenotype data have genotypes, but there are 33 genotyped mice with no phenotypes, including 3 Gough mice and 30 F2 progeny.

Reproducible reports

Gough project diagnostics

Karl

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Ther
data
mice

```
I've combined the initial genotypes (using the re-clustered genotypes
for plates 14-16) with the well-behaved portion of the re-run
genotypes. I'm focusing on `r totmar(g)` markers that are informative
(though, as we'll see, there are still a lot of badly behaved and
basically non-informative markers that need to be removed).
I've combined data on replicate samples, to give one set of genotype
calls for each sample.

There are `r nind(g)` genotyped mice and `r nrow(phe)` phenotyped
mice. All of the mice in the phenotype data have genotypes, but there
are `r sum(is.na(match(gid, pid)))` genotyped mice with no phenotypes,
including `r sum(g$pheno$gen[which(is.na(match(gid, pid)))]==0)`
Gough mice and `r sum(g$pheno$gen[which(is.na(match(gid, pid)))]==2)`
F2 progeny.
```

Automate the process (GNU Make)

```
R/analysis.html: R/analysis.Rmd Data/cleandata.csv
    cd R;R -e "rmarkdown::render('analysis.Rmd')"

Data/cleandata.csv: R/prepData.R RawData/rawdata.csv
    cd R;R CMD BATCH prepData.R

RawData/rawdata.csv: Python/xls2csv.py RawData/rawdata.xls
    Python/xls2csv.py RawData/rawdata.xls > RawData/rawdata.csv
```

Automate the process (GNU Make)

```
R/analysis.html: R/analysis.Rmd Data/cleandata.csv
    cd R;R -e "rmarkdown::render('analysis.Rmd')"

Data/cleandata.csv: R/prepData.R RawData/rawdata.csv
    cd R;R CMD BATCH prepData.R

RawData/rawdata.csv: Python/xls2csv.py RawData/rawdata.xls
    Python/xls2csv.py RawData/rawdata.xls > RawData/rawdata.csv
```

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```
R/analysis.html: R/analysis.Rmd Data/cleandata.csv
    cd R;R -e "rmarkdown::render('analysis.Rmd')"

Data/cleandata.csv: R/prepData.R RawData/rawdata.csv
    cd R;R CMD BATCH prepData.R

RawData/rawdata.csv: Python/xls2csv.py RawData/rawdata.xls
    Python/xls2csv.py RawData/rawdata.xls > RawData/rawdata.csv
```

Automate the process (GNU Make)

```
R/analysis.html: R/analysis.Rmd Data/cleandata.csv
    cd R;R -e "rmarkdown::render('analysis.Rmd')"

Data/cleandata.csv: R/prepData.R RawData/rawdata.csv
    cd R;R CMD BATCH prepData.R

RawData/rawdata.csv: Python/xls2csv.py RawData/rawdata.xls
    Python/xls2csv.py RawData/rawdata.xls > RawData/rawdata.csv
```

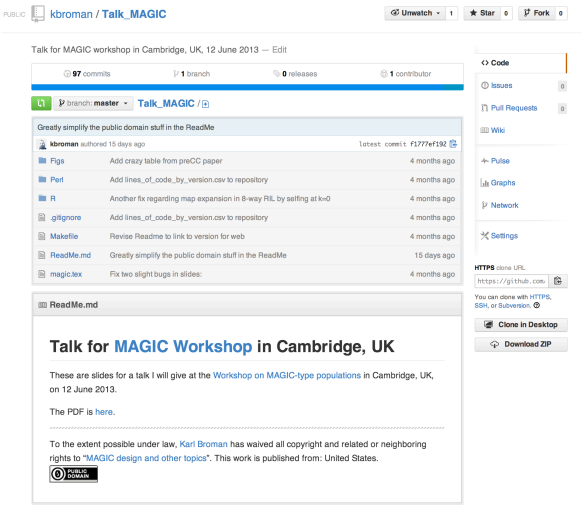
Write modular code

- Modular code is easier to understand, maintain, and reuse.
- Turn repeated code into functions
- Combine useful functions into a package or module

Keeping track of versions

- Google drive / Dropbox / Box
- Version numbers in file names
- Formal version control (e.g., git/GitHub)
  - Browse changes
  - Try new things without fear of breaking what works
  - Jump to the state of the project at any time point
  - Merge simultaneous changes from multiple people

Version control (git/GitHub)



Version control (git/GitHub)

PUBLIC kbroman / Talk_MAGIC

Unwatch 1 | Star 0 | Fork 0

Talk for MAGIC workshop in Cambridge, UK, 12 June 2013 — Edit

97 commits | 1 branch | 0 releases | 1 contributor

Code

Issues 0

Greatly simplify the public domain stuff in the ReadMe

kbroman authored 15 days ago — latest commit f1777ef192

Figs	Add crazy table from preCC paper	4 months ago
Perl	Add lines_of_code_by_version.csv to repository	4 months ago
R	Another fix regarding map expansion in 8-way RIL by selfing at k=0	4 months ago
.gitignore	Add lines_of_code_by_version.csv to repository	4 months ago
Makefile	Revise Readme to link to version for web	4 months ago
ReadMe.md	Greatly simplify the public domain stuff in the ReadMe	15 days ago
magic.tex	Fix two slight bugs in slides:	4 months ago

rights to "MAGIC design and other topics". This work is published from: United States.

PUBLIC DOMAIN

Version control (git/GitHub)

PUBLIC kbroman / Talk_MAGIC

Unwatch 1 | Star 0 | Fork 0

branch: master Talk_MAGIC / Commits

Sep 27, 2013

- **Greatly simplify the public domain stuff in the ReadMe** — kbroman authored 15 days ago — f1777ef192 — Browse code
- **Fix url in ReadMe.md file** — kbroman authored 15 days ago — a65158273f9 — Browse code

Jun 17, 2013

- **Another fix regarding map expansion in 8-way RIL by selfing at k=0** — kbroman authored 4 months ago — 20aa482f2c — Browse code
- **Fix two slight bugs in slides:** HEAD — kbroman authored 4 months ago — 51d4aa9ceb — Browse code

```
- 8-way RIL by selfing: map expansion = 1 at k=0

- Slight repair to definition of 3-pt coincidence
```

Jun 10, 2013

- **Change one page number** — kbroman authored 4 months ago — e8e0608615 — Browse code
- **Add missing paren** — kbroman authored 4 months ago — f4975dee6e — Browse code
- **who's -> who is** — kbroman authored 4 months ago — 886f20f098 — Browse code
- **rubbish -> bad** — kbroman authored 4 months ago — e6fbf2f647 — Browse code
- **Add link to R/qtl page** — kbroman authored 4 months ago — 4edf3e8d76 — Browse code
- **Revise slide re analysis issues** — kbroman authored 4 months ago — 14ebb1eeb5 — Browse code
- **italicize 'de novo'** — kbroman authored 4 months ago — 45dda4b4c7 — Browse code
- **replace plain right arrow with fat arrow** — kbroman authored 4 months ago — 8bbe305d6c — Browse code

Version control (git/GitHub)

PUBLIC kbroman / Talk_MAGIC

Unwatch 1 | Star 0 | Fork 0

Fix two slight bugs in slides:

Browse code

```
- 8-way RIL by selfing: map expansion = 1 at k=0

- Slight repair to definition of 3-pt coincidence
```

master

kbroman authored 4 months ago — 1 parent e0e0608 commit 51d4aa9ceb104bbf26e0cbe105a5c7f8dc02a832

Showing **2 changed files** with **5 additions** and **3 deletions**. Show Diff Stats

6 R/map_expansion_func.R — View file @ 51d4aa9

```diff
@@ -25,8 +25,10 @@ mesibA4 <- function(k)
 ###############
 # Eight-way
 ###############
-meself8 <- function(k)
-    4 - (((1)/(2)))^(k-2)
+meself8 <- function(k) {
+  if(k==0) return(1)
+  4 - (((1)/(2)))^(k-2)
+}

 mesibX8 <- function(k)
     ((14)/(3)) - (((30 + 14*sqrt(5))/(15))) * (((1+sqrt(5))/(4)))^k - (((30 - 14*sqrt(5))/(15))) * (((1-sq
```

2 magic.tex — View file @ 51d4aa9

```diff
@@ -636,7 +636,7 @@
 \hspace{20mm} {\color{myblue} = $\mathsf{Pr(\text{rec'n in 23} \ |
    \ \text{rec'n in 12}) /
-  Pr(\text{rec'n in 12})}$}
+  Pr(\text{rec'n in 23})}$}

 \item
 No interference { \color{myblue} = 1 }
```

Version control (git/GitHub)

```
 ################
-meself8 <- function(k)
-    4 - (((1)/(2)))^(k-2)
+meself8 <- function(k) {
+  if(k==0) return(1)
+  4 - (((1)/(2)))^(k-2)
+}

 mesibX8 <- function(k)
     ((14)/(3)) - (((30 + 14*sqrt(5))/(15)))
```

2 magic.tex

```
@@ -636,7 +636,7 @@

 \hspace{20mm} {\color{myblue} = $\mathsf{Pr(
     \ \text{rec'n in 12}) /
-  Pr(\text{rec'n in 12})}$}
+  Pr(\text{rec'n in 23})}$}

```

Backups

- Multiple places, including off-site
- Automatic

License your software

Pick a license, any license

– Jeff Atwood

Share your stuff

- Code
  - GitHub / BitBucket
  - Zenodo (archival, with DOIs)
- Data
  - Domain-specific repository (e.g., dbGAP)
  - General repository (e.g., github, figshare, zenodo, datadryad)
  - Institutional repository

Summary

1. Organize your project
2. Choose good names for things
3. Document what's what
4. Organize data as a rectangle
5. Metadata is data
6. Everything with a script
7. Even better: reproducible reports
8. Automate the process (GNU Make)
9. Write modular code (functions and packages)
10. Use version control (git/GitHub)
11. License your software
12. Share your data and code

Other considerations

- Testing
  - are you getting the right answers?
- Software versions
  - will your stuff work when dependencies change?
- Large-scale computations
  - computation time + dependence on cluster environment
- Collaborations
  - coordinating who does what and where things live

It's not that we don't test our code,

it's that we don't store our tests
so they can be re-run automatically.

– Hadley Wickham

R Journal 3(1):5–10, 2011

Testing and debugging

find a bug $\longrightarrow$ write a test

Testing and debugging

find a bug $\longrightarrow$ write a test

fix a bug $\longrightarrow$ look for other instances of that mistake

Collaboration

- Do more, by working in parallel
- Do more, through diversity of ideas and skills
- Reproducible pipelines have immediate advantages
- Tests of reproducibility
- Code review

Challenges in collaborations

- Shared vision?
- Compromise
- Coordination
- Communication
- Sharing code and data
- Synchronization

Challenges in collaborations

- Shared vision?
- Compromise
- Coordination
- Communication
- Sharing code and data
- Synchronization
- Weakest link?

Challenges

(totally hypothetical)

“Could we meet to talk about the data file structure?”

“Could we meet to talk about the data file structure?”

“No.”

“Wait, these results seem to be based on the older SNP map.”

“Could you write the methods section?”

“But I didn’t do the work,
and we don’t have the code that was used.”

“My data analyst has taken a job at Google.”

“Could you do these analyses? X said they would, but they’re not responding to my emails.”

Shared vision

- Publication
- Code & data sharing
- Who will do what
- Timeline
- Ongoing sharing of methods, results

Shared workspace

- Project structure
- Data and metadata formats
- Software environment
- Automated sync (or it won't happen)

Technology for sharing

- Data
  - figshare
  - dropbox / box / google drive
- Code
  - github / bitbucket
- Pipeline / workflow
  - make / drake / snakemake / rake
- Full environment
  - docker containers
  - `mybinder.org` / `wholetale.org`

The most important tool is the mindset,
when starting, that the end product
will be reproducible.

– Keith Baggerly

The second-most important tool is training.

– me

Slides: kbroman.org/Talk_ReproRes

PUBLIC DOMAIN

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